

AI magic in ASD management

- Accurate and timely diagnosis
- Data-driven insights
- Personalised and adaptive interventions
- Increased engagement and motivation, due to personalisation
- Better communication and social interaction
- Greater support for caregivers and therapists
- Accessibility and scalability of interventions, in underserved regions too

seamlessCARE's Empathic is an app that can interpret the emotions of non-verbal people. Usually, the primary caregivers can adeptly understand what a non-verbal person is trying to express. Still, not everyone in the extended circle of caregiving might be able to do so. Empathic app comes in handy at such times. It interprets emotional states from facial expressions, voice tones, and physiological responses, allowing caregivers and therapists to understand better and respond to an individual's emotional needs and triggers. iMotions Lab also offers customised hardware and software solutions for understanding and interpreting emotions using not just video but also physiological parameters.

Tutorials for parents and caregivers. AI-driven apps can offer guided step-by-step tutorials for parents and suggest tailored activities. SocialMind, for example, is a smartphone app that trains caregivers and therapists on strategies to motivate children with autism to talk and communicate. The app first teaches the caregiver a technique. Then, it records the video of the caregiver implementing the recommended technique, analyses the implementation and the child's response, and gives personalised recommendations to improve the technique, or takes you to the next level with further recommendations.

Predictive analytics. AI can predict potential challenges or behavioural episodes by analysing

past triggers and patterns, enabling proactive management and early intervention. Empatica, for example, uses its Embrace-Plus smartwatch and associated platform to anticipate and potentially prevent aggressive outbursts by warning caregivers on time. The wearable device monitors physiological data, such as heart rate, skin conductance, and movement, which are then analysed on the platform to identify patterns preceding such events.

Personalised interventions

One size does not fit all—true for anything and everything, but especially true for individuals on the autistic spectrum. ASD includes a wide range of social, communication, and behavioural challenges, and each challenge can be mild, severe, or somewhere in between. So, how can one standardised solution fit everyone? With or without AI, therapists and caregivers have always been customising solutions and devising new strategies on the fly for individuals with different symptoms. But now, modern AI tools have made personalised

interventions easier and more structured too, based on scientific data analysis and real-time tracking.

Enabling better communication. Augmentative and alternative communication (AAC) devices are communication tools for those who cannot rely on speech. Therapists may recommend various types of AAC devices based on the nature and severity of the disability. No-tech options include gestures (could even be a simple head tilt or blink), eye movements, and more. Low-tech options include chalkboard, picture cards, and such. High-tech options include tablets with special apps, speech recognition systems, and emerging AI-powered tools.

Where possible, the therapist might encourage no-tech options, at least for essential communications, to reduce dependency on a gadget, which can fail at any time, leaving the person helpless. Although the needs and applications may vary, neurodivergent individuals find AAC devices very helpful. Many tech startups and ASD research foundations are starting to integrate the power of AI into AAC tools to enable those on the autism spectrum to communicate better, without getting anxious. AI enhances AAC devices with features like personalised learning, predictive text, and voice recognition.

Advancements in natural language processing (NLP) and AI have given rise to personalised speech therapy apps that provide

To-do list for the tech and research community

- Affordable, accessible, and more efficient AI models
- Ethical, cultural, and linguistic compatibility of AI apps
- Comfortable, neurodivergence-friendly wearables
- Collaborative development with researchers, caregivers, educators, and therapists
- Comprehensive, longitudinal research to establish the long-term efficacy of AI-powered tools
- Neurodivergence-friendly features in common apps and gadgets
- Regulatory frameworks to ensure data privacy, security, and ethical use